



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
REVIEW TEST
02-01-2022

JEE(Main + Advanced) : NURTURE COURSE [PHASE : TOPS, I(A), I, I(B), II(A), II & LIVE-I, I(B), II(A)]

Time : 3 Hours

PAPER-1

Maximum Marks : 198

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16,
Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24,
Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127,
Xe = 131, Ba = 137, Ce = 140, Cu = 63.5, Ne = 20, K = 39, Mn = 55

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks: 18)

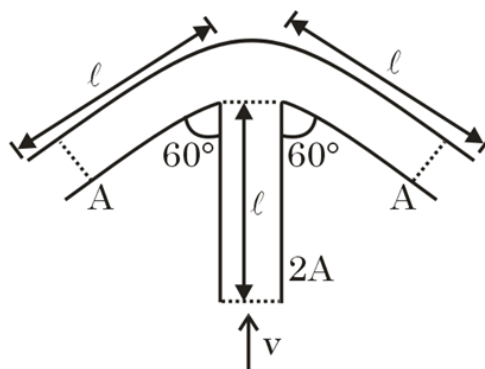
- This section contains **SIX (06)** questions.
- Each question has Four options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. An ideal liquid of density ρ is pushed with velocity v through the central limb of the tube shown in the figure. The cross-sectional areas is as shown figure and length of each arm is equal to ℓ . Assume stream-line flow. Find minimum friction coefficient required between pipe and surface such that pipe doesn't move. (Assume pipe is massless and volume of liquid is along length of three pipes)



(A) $\frac{3}{2} \frac{v^2}{g\ell}$

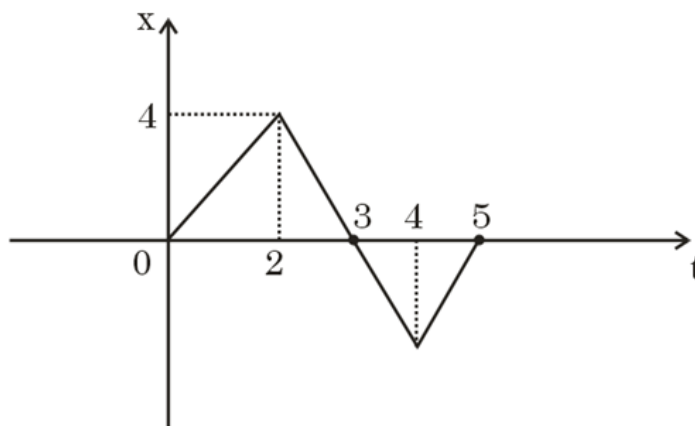
(B) $\frac{v^2}{g\ell}$

(C) $\frac{5}{4} \frac{v^2}{g\ell}$

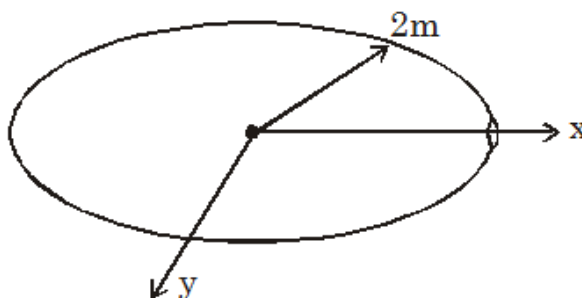
(D) $\frac{3}{4} \frac{V^2}{g\ell}$

Space for Rough Work

2. Position-time graph of a particle of mass 2kg is shown in figure. Then from $t = 0$ to $t = 5$ s :



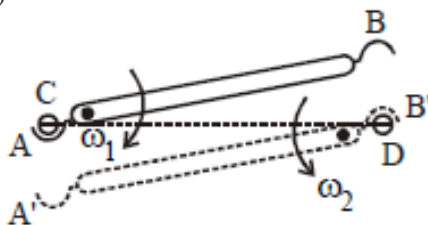
- (A) Displacement of particle is non-zero. (B) Work done on particle is zero.
 (C) Impulse on particle is zero. (D) Work done on particle is 12J.
3. A bead of mass 300 gm is threaded on a smooth horizontal circular wire frame of radius 2m. The centre of frame is at the origin and frame is in xy plane. The potential energy of the bead due to some unknown field is $U = x + 2y^2$. The bead is at $x = +2$ m and its total energy is 4J. The normal force exerted by wire frame is (Take : $g = 10 \text{ m/s}^2$ in downward direction).



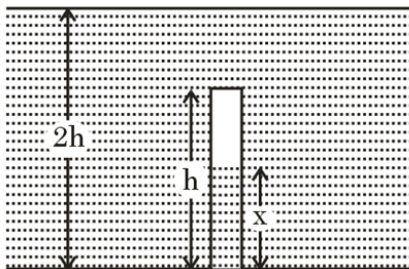
- (A) $\sqrt{10} \text{ N}$ (B) 4 N (C) 0 N (D) 5 N

Space for Rough Work

4. A uniform thin rod AB of length L is equipped at both ends with the hooks shown and is supported by a frictionless horizontal table. Initially the rod is hooked at A to a fixed pin C about which it rotates with a constant angular velocity ω_1 . Suddenly end B of the rod hits pin D and gets hooked to pin D, causing end A to be released. Find Impulse at point D (Assume length and mass of the hook is negligible. Pin C & D are lying on a same horizontal line)



- (A) $3ML\omega_1$ (B) $\frac{ML\omega_1}{4}$ (C) $ML\omega_1$ (D) None of these
5. A tube of length h (which is wide enough so that surface tension effects can be neglected) filled with air at atmospheric pressure and closed at one end. Now the tube is lowered in a tank of mercury to a depth 2h as shown. It is seen that mercury rises a distance x into the tube. If the mercury barometer also reads h, then (Consider compression of air to be Isothermal)



- (A) $(3h - x)(h - x) = h^2$ (B) $(2h - x)(h - x) = h^2$
 (C) $(2h - x)(h + x) = h^2$ (D) $(2h)(h - x) = h^2$
6. A thin glass vessel ($\alpha_g = \frac{5}{2} \times 10^{-5}/^\circ\text{C}$) is partially filled with a liquid ($\gamma_l = 1.3 \times 10^{-5}/^\circ\text{C}$) at 20°C . If this vessel is heated to 40°C ; the level of liquid in the vessel will :-
- (A) Increase (B) Decrease
 (C) Remains same (D) None of the above

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

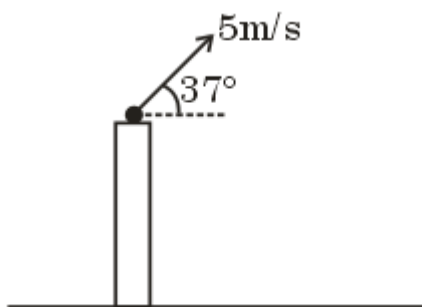
Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

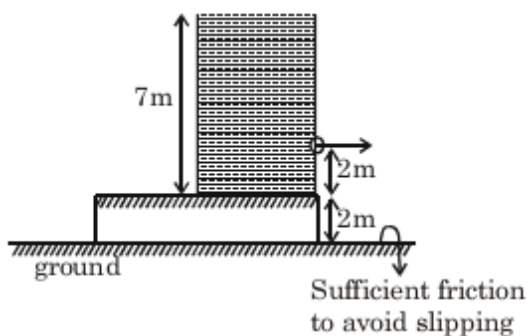
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7. A metal cylinder of mass 0.5 kg is heated electrically by a 12 W heater in a room at 15°C. The cylinder temperature rises uniformly to 25°C in 5 min and finally becomes constant at 45°C. Assuming that the rate of heat loss is proportional to the excess temperature over the surroundings,
- (A) the rate of loss of heat of the cylinder to surrounding at 20°C is 2W
- (B) the rate of loss of heat of the cylinder to surrounding at 45°C is 12W
- (C) the rate of loss of heat of the cylinder to surrounding at 20°C is 5W.
- (D) the rate of loss of heat of the cylinder to surrounding at 45°C is 30W.
-

Space for Rough Work

8. A ball is thrown from the top of 36 m high tower and wind is giving constant acceleration of 10 m/s^2 in left direction then



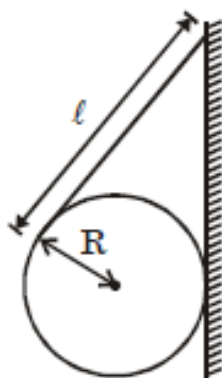
- (A) Ball will land on left side of tower at 33 m from its foot
 (B) Ball will land on tower 8 m below from point of projection
 (C) The minimum velocity during flight is 4 m/s
 (D) The minimum velocity of ball during the flight is 0.35 sec after projection.
9. A cylindrical vessel is full of water (density $\rho = 1 \text{ gm/cc}$) is placed on horizontal ground as shown. A small hole of area ($= 1 \text{ cm}^2$) is made on a curved wall at a height of 2m from the bottom of container (container does not move) choose **CORRECT** statement(s) ($g = 10 \text{ m/s}^2$) :-



- (A) Horizontal range of water jet is maximum when water level is at height of 8 m from ground
 (B) Maximum horizontal force on container by table is 10 Newton
 (C) Maximum horizontal range of water jet on ground is $4\sqrt{5}$ meter
 (D) Horizontal range of water jet will decrease then increase with decrease of water level

Space for Rough Work

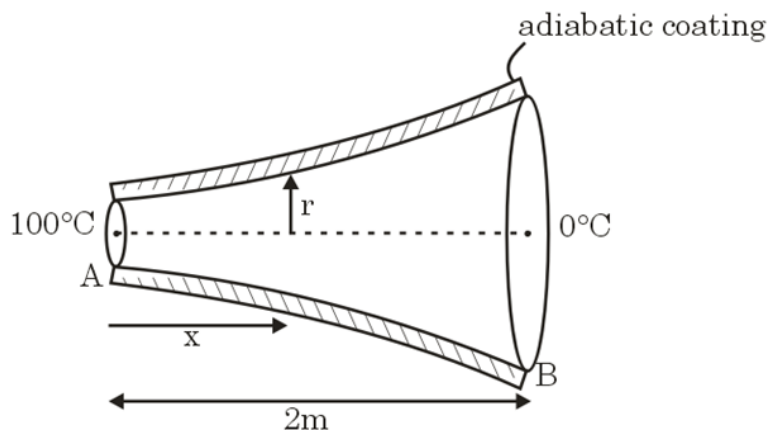
10. Spool of thread having radius R is tied to a wall as shown. The mass of the thread is negligible and mass of spool is M . The thread is along the tangent to the spool.



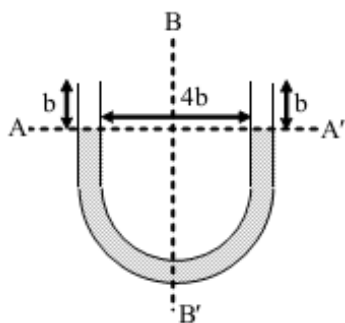
- (A) The tension in the thread is $\frac{Mg}{2} \left(1 + \frac{R^2}{\ell^2} \right)$.
- (B) The normal force on the wall is $\frac{MgR}{2\ell}$.
- (C) The friction coefficient can be $\frac{R^2 + \ell^2}{4\ell R}$.
- (D) The friction coefficient can be $\frac{R^2 + \ell^2}{3\ell R}$.

Space for Rough Work

11. Radius of a metallic rod varies with x (distance from end A) as $r = (x + 1)^{3/2}$ m. End A and B of rod are maintain at 100°C and 0°C respectively. Thermal conductivity and length of rod are $\frac{3}{\pi}$ W/m-K and 2 m respectively. Assume steady state flow of heat in rod and rod is covered with adiabatic coating :-



- (A) Rate of heat flow through the rod is 675 W
 (B) Value of x at which temperature is 50°C , is 34.1 cm
 (C) Temperature is approximately 15.6°C at $x = 1$ m
 (D) Ratio of thermal resistance of rod between ($x = 0$ & $x = 1$) and ($x = 1$ & $x = 2$) is $\frac{27}{5}$
12. A tube is filled with a liquid as shown in figure. Then –



- (A) Liquid will spill out left tube if the tube is accelerated towards right horizontal direction with $a > g/2$
 (B) Liquid will spill out right tube if tube is rotated about left arm with angular velocity $\omega > \sqrt{\frac{g}{4b}}$
 (C) Liquid will not spill out the tube if it is rotated about axis AA' no matter what angular speed is
 (D) Liquid will spill out the tube if it is rotated about axis BB' with angular velocity $\omega > \sqrt{\frac{g}{2b}}$

Space for Rough Work

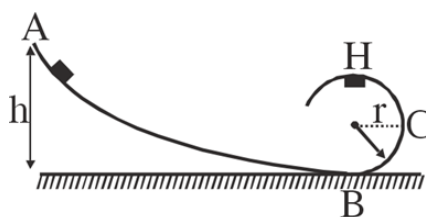
SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
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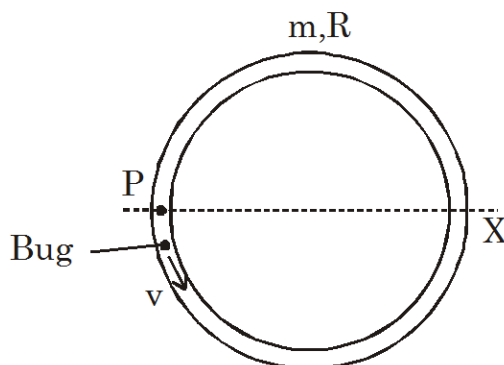
Zero Marks : 0 In all other cases.

1. A parallel beam of nitrogen molecules moving with velocity $v = 400$ m/s impinges on a wall at an angle of 30° to its normal. The concentration of molecule in the beam $n_0 = 6.02 \times 10^{19} \text{ cm}^{-3}$. If the pressure exerted by the beam on the wall assuming the collisions of molecules with the wall are perfectly elastic is $\frac{n}{2} \times 10^{-7}$ (Pa). Then value of n is
2. A sailor in a boat, which is going due east with a speed of 8 m/s, observes that a submarine is heading towards north at a speed of 12 m/s and sinking at a rate of 2 m/s. The commander of submarine observes a helicopter ascending at a rate of 5 m/s and heading towards west with 4 m/s. Find the speed of the helicopter with respect to boat (in m/s).
3. A small block of mass m is allowed to slide from $h = 2r$ on an inclined frictionless track in vertical plane from rest position as shown in the figure. Find total acceleration of block at point C is



Space for Rough Work

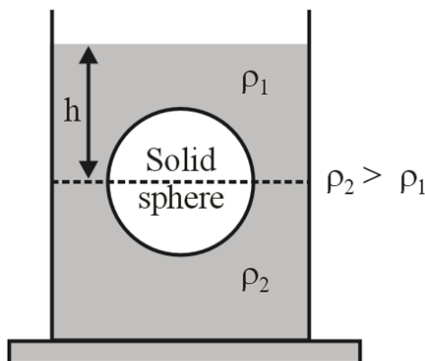
4. A thin ring of mass (100 gm) and radius R (3 m) is pivoted at point P. A bug of mass (50 gm) starts moving from point P along the circumference of ring with speed V (8 m/s) w.r.t to ring. Find speed (in m/s) of bug w.r.t ground when it reaches at diametrically opposite point X of point P. Bug is moving with constant speed w.r.t. ring.



5. A block of metal weighing 2 kg is resting on a frictionless plane. It is struck by a jet releasing water at a rate of 1 kg/s and at a speed of 5 m/s. The initial acceleration of the block in (m/s^2) is



6. A solid sphere of radius r is floating at the interface of two immiscible liquids of densities ρ_1 and ρ_2 ($\rho_2 > \rho_1$), half of its volume lying in each. The height of the upper liquid column from the interface of the two liquids is h . The force exerted on the sphere by the upper liquid is $a\rho_0\pi r^2 + (h - br)\pi r^2\rho_1g$ then value of $a + b$ is



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I(i) : (Maximum Marks: 18)

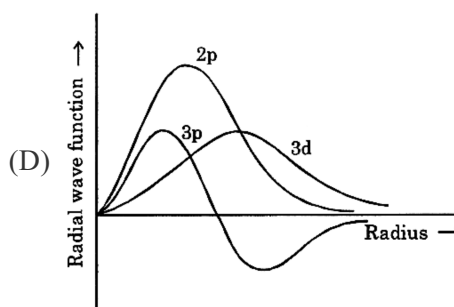
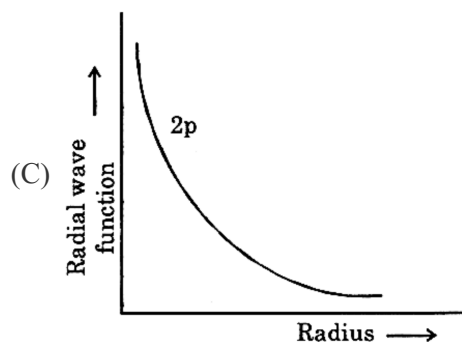
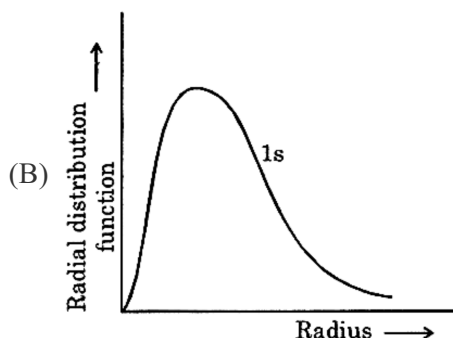
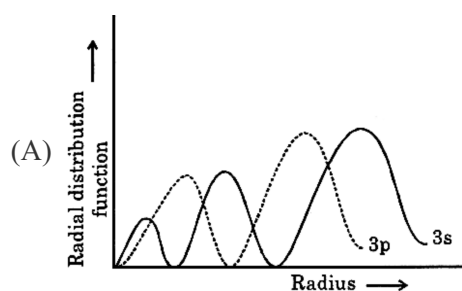
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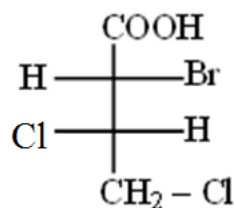
Negative Marks : -1 In all other cases

- When N_2O_5 is heated at temp. T, it dissociated as $\text{N}_2\text{O}_5 \rightleftharpoons \text{N}_2\text{O}_3 + \text{O}_2$, $K_c = 2.5$. At the same time N_2O_3 also decomposes as : $\text{N}_2\text{O}_3 \rightleftharpoons \text{N}_2\text{O} + \text{O}_2$. If initially 4 moles of N_2O_5 are taken in 1.0 litre flask and allowed to attain equilibrium, concentration of O_2 was formed to be 2.5M. Equilibrium concentration of N_2O is :-
(A) 1.0 M (B) 1.5 M (C) 2.16 M (D) 0.33 M
- Which of the following plot is incorrect?



Space for Rough Work

3. The specie among the following that has maximum difference of number of BONDS of central atom between Lewis structure and structure according to VSEPR is.
- (A) H_2SO_4 (B) POF_3 (C) CO_3^{2-} (D) ClO_4^-
4. The pair in which 1st atom has more EA than 2nd atom is.
- (A) O(g), S(g) (B) F(g), Cl(g) (C) P(g), N(g) (D) I(g), Br(g)
5. Total number of structural isomers are possible with molecular formula $\text{C}_4\text{H}_{10}\text{O}$ are :
- (A) 5 (B) 6 (C) 7 (D) 8
6. Assign R, S configuration for given compounds.



- (A) 2R, 3S (B) 2S, 3S (C) 2S, 3R (D) 2R, 3R

Space for Rough Work

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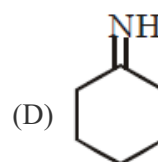
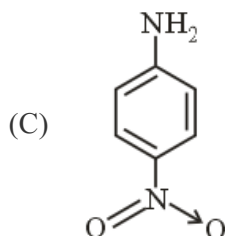
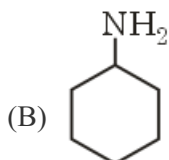
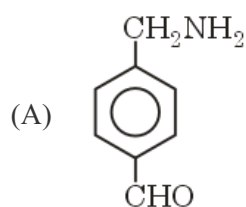
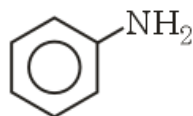
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7. A given solution of H_2SO_4 is labelled as 49% (w/w), then correct statement regarding the solution is ($d = 1.3 \text{ g/ml}$)
- (A) Molality = $\frac{500}{51}$ (B) Normality = $\frac{1000}{51}$
- (C) % w/v = $(49 \times 1.3)\%$ (D) Molarity = 6.5
8. Identify correct statement (s)
- (A) Critical temperature of a gas is the highest temperature at which liquefaction of gas first occurs
- (B) The densities of liquid and gas is same at critical condition
- (C) As the value of Vander waal's constant 'a' increases, the gas becomes difficult to liquify
- (D) A sample of gas in which attractive forces are more dominant is expected to have a high value of critical temperature
9. The maximum number of electrons in Ti^{2+} for which $\ell \times m \leq -1$ is.
- (A) 4 (B) 5 (C) 6 (D) 7
-

Space for Rough Work

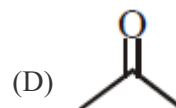
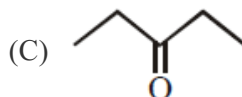
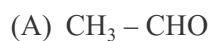
10. The oxo-acids that will have atleast one oxo-linkage is.

- (A) $K_2Cr_2O_7$ (B) Cl_2O_7 (C) $H_5P_3O_{10}$ (D) $H_2S_2O_7$

11. Which of the following compound(s) have C–N bond length more than C–N bond length in Aniline ?



12. Select the compound(s) whose enol form show(s) geometrical isomerism?



Space for Rough Work

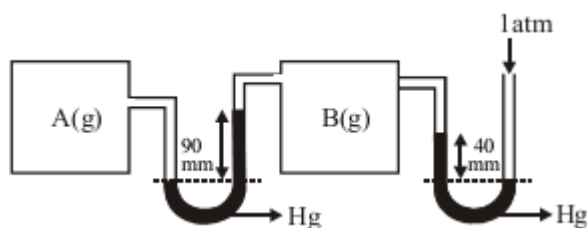
SECTION-II : (Maximum Marks: 24)

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Zero Marks : 0 In all other cases.

1. At 300K, two gases are filled in two equal sized containers as given.

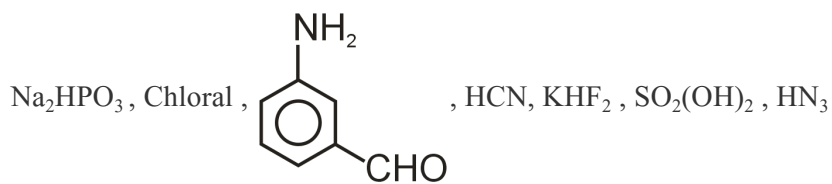


What will be the pressure of A(g) (in mm of Hg).

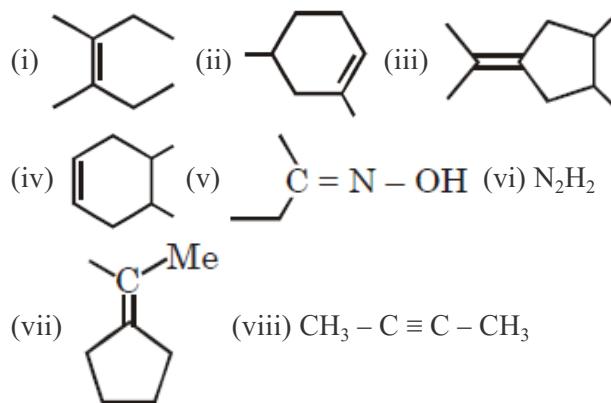
2. 1 litre solution containing 0.1 M $\text{Na}_2\text{S}_2\text{O}_3$ & 0.2 M IO_4^- is titrated with I_2 according to following reaction.
- $$\text{Na}_2\text{S}_2\text{O}_3 + \text{I}_2 \rightarrow \text{Na}_2\text{S}_4\text{O}_6 + \text{I}^-$$
- $$\text{IO}_4^- + \text{I}_2 \rightarrow \text{IO}_3^-$$
- Calculate millimoles of I_2 consumed ?
3. If x = number of π bonds in sodium peroxoborate
 y = number of atoms in 3d series having more than 3 unpaired e^- s
 z = number of 'd' orbitals having two nodal planes
 then $(x + z - y)$ is _____.

Space for Rough Work

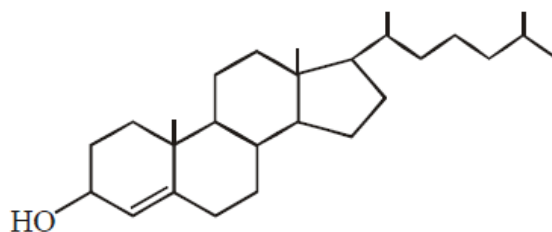
4. The number of specie(s) that do not show hydrogen bonding among the following.



5. How many compound(s) can show G.I.?



6. Calculate total number of chiral center present in following compound



Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I(i) : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- Each question has Four options (A), (B), (C) and (D). ONLY ONE of these four options is correct.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. If a,b,c are positive numbers which are in A.P., then the sum of coefficients when $(-1 + (ax^2 - 2bx + c))^{2015}$ is expanded in powers of x is equal to -
 (A) 0 (B) 1 (C) -1 (D) 2^{2015}
2. The number of positive integral solutions of $\frac{(2 - x^2)(x - 3)^3}{(x + 1)(x^2 - 3x - 4)} \geq 0$, is -
 (A) 3 (B) 2 (C) 4 (D) 0
3. Consider a parabola whose focus is (1, 2) and its directrix is common tangent of ellipse $\frac{x^2}{4} + \frac{y^2}{1} = 1$ and $\frac{x^2}{1} + \frac{y^2}{4} = 1$. If x & y intercept of common tangent are positive. Length of latus rectum of parabola is -
 (A) $(3 - \sqrt{5})$ (B) $(3 + \sqrt{5})$ (C) $\sqrt{2}(3 - \sqrt{5})$ (D) $\sqrt{2}(3 + \sqrt{5})$

Space for Rough Work

4. If $a_1, a_2, \dots, a_{300}$ are in Arithmetic progression such that $\sum_{n=1}^{100} a_{3n} = 200$ & $\sum_{n=1}^{100} a_{3n-2} = 100$, then a_1 is -
(A) $-\frac{293}{4}$ (B) $-\frac{297}{4}$ (C) -73 (D) -74
5. Sum of all the solution(s) of equation $9^{\log_3(\log_2 x)} = \log_2 x - \log_2^2 x + 1$ is -
(A) a prime number (B) an irrational number
(C) a rational number which is not integer (D) a composite number
6. Which of the following is true ?
(A) $\sin 1 > \sin 2 > \sin 5$ (B) $\sin 5 > \sin 2 > \sin 1$
(C) $\sin 2 > \sin 1 > \sin 5$ (D) $\sin 1 > \sin 5 > \sin 2$
-

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

-
7. Let $x = (15 + \sqrt{224})^n = I + F$, where I & F are integral and fractional part of x respectively. Then (where $n \in \mathbb{N}$)
- (A) $x(1 - F) = 1$ (B) $x(1 - F) = 2^n$ (C) I is odd (D) I is even
8. 6 digit numbers are formed with digits 1, 2, 3, 4, 5, 6 (without repetition). Let the number of ways are λ in which both middle digits are odd, then λ is divisible by -
- (A) 2 (B) 4 (C) 12 (D) 72
9. Let $(a - 2)x^2 + (b - 3)x + (c - 4) = 0$ has three distinct real roots, then
- (A) $a + b + c = 9$ (B) $a^2 + b^2 - c = 9$
- (C) $a + b + 2c = 13$ (D) $\frac{a + b + c}{c} = 2$
-

Space for Rough Work

10. Number of real value(s) of c for which system of equations $x^2 + cx - 1 = 0$ and $cx^2 + x - 1 = 0$ have
- (A) a common root, is 1 (B) two common roots, is 1
(C) exactly one common solution, is 1 (D) a common root, is 2
11. If $(\log_x 2 + \log_y 2 + \log_z 2)^2 = (\log_x 2)^2 + (\log_y 2)^2 + (\log_z 2)^2$, then
- (A) $\log_x 2 + \log_y 2 + \log_z 2 = 0$ (B) $xyz = 1$
(C) $\frac{1}{\log_x 2} + \frac{1}{\log_y 2} + \frac{1}{\log_z 2} = 0$ (D) $xyz = 2$
12. Let N be the number of values of x for which $\frac{\tan 3x + \tan(-2x)}{1 + \tan 3x \tan 2x} = 1$, $x \in [0, 4\pi]$, then
- (A) $N > 3$ (B) $N < 6$ (C) $N > 4$ (D) $N < 4$

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SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. Let coefficient of x^2 and x^3 in the expansion of $(3 + ax)^9$ are equal. If 'a' lies between the roots of the equation $49x^2 + 7px - 9 = 0$, then the range of p is $(-\infty, -K)$. Then the value of K is
2. Value of $\frac{\sin 32^\circ + \sin 28^\circ - \sin 60^\circ}{\sin 16^\circ \cdot \sin 14^\circ \cdot \sin 30^\circ}$ is equal to
3. If α, β, γ are the 3 positive roots of cubic $x^3 - ax^2 + bx - 16\sqrt{6} = 0$, then the minimum value of $\left(\frac{4}{\alpha\beta} + \frac{3}{\beta\gamma} + \frac{2}{\gamma\alpha}\right)$ is -
4. Let $P(x)$ be a polynomial such that when $P(x)$ is divided by $(x - 11)$, then the remainder is 53 and when $P(x)$ is divided by $(x - 99)$, then the remainder is -35. If $r(x)$ is the remainder when $P(x)$ is divided by $(x - 11)(x - 99)$ then $\frac{r(63)}{2}$ is equal to
5. Given $A(1,1)$ and $B(2,3)$ are two vertices of a triangle ABC and interior angle bisector of angle A is $x - 2y + 1 = 0$. If 'm' be the slope of AC and $|m| = \frac{a}{b}$ where a and b are coprime, then $\frac{(b - a)}{2}$ is equal to
6. If $1 \cdot ({}^{20}C_0)^2 + 2 \cdot ({}^{20}C_2)^2 + 3 \cdot ({}^{20}C_4)^2 + \dots + 11 \cdot ({}^{20}C_{20})^2$ is equal to $3 [{}^{8a}C_{4a} + {}^{4a}C_{2a}]$, then a is equal to

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Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them._____
Signature of the CandidateI have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same._____
Signature of the Invigilator

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